Experiment 07:

Name of the Experiment: MOSFET Amplifier

Objective : Become familiar with the MOSFET amplifier.

Theory:

MOSFET is a three terminal electronic device. It is a voltage controlled device. There are two types of MOSFET : 1) Depletion type MOSFET and 2) Enhancement type MOSFET. IRF 540 is the n-channel enhancement type MOSFET. The symbol of this MOSFET is shown in Fig.1

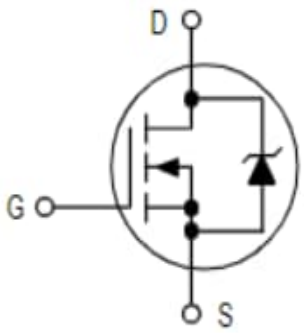


Fig.1 *n-channel enhancement-type MOSFET*

The terminals of MOSFET are: Gate (G), Drain (D), and Source (S). For an n-channel Enhancement-type MOSFET when gate to source voltage, $V_{GS} > V_{th}$ the drain current, I_D is related to the V_{GS} by the following equation:

$$I_D = k(V_{GS} - V_{th})^2$$

where
I_D is the drain current.
The V_{GS} is the Gate to Source Voltage.
The V_{th} is the threshold voltage at which the MOSFET turns on
k is the transconductance parameter which solely depends on the type MOSFET being used.

Biasing Techniques

To set a proper operating point (Q-point) for MOSFET circuits, several biasing techniques are used. Here voltage divider biasing is discussed.

Voltage Divider Biasing Circuit: The circuit given in Fig.2, shows how the voltage divider circuit works. What this circuit does is that since $I_G = 0\ A$, it takes the voltage V_{DD} and divides it amongst R_1

and R_2, thus, using Voltage Divider Rule, the gate voltage (V_G) applied will be equal to the voltage across R_2.

$$V_G = \frac{R_2}{R_1 + R_2} V_{DD}$$

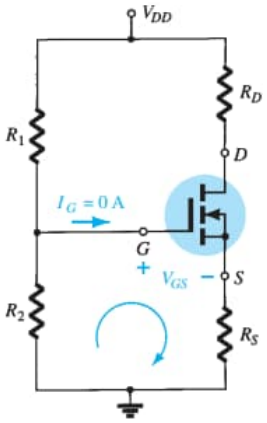


Fig.2

Using KVL in the given loop in Figure 7.3, we can get the following equation:

$$V_{GS} = V_G - I_D R_S$$

Recall that drain current (I_D) flowing through R_D and R_S are the same.

Transfer Curve

The Transfer Curve for an n-type MOSFET can be found using Shockley's equation, the graph in Fig.3 shows a sudden exponential increase of drain current when the gate-to-source voltage applied is greater than the threshold voltage $(V_{GS} > V_{th})$.

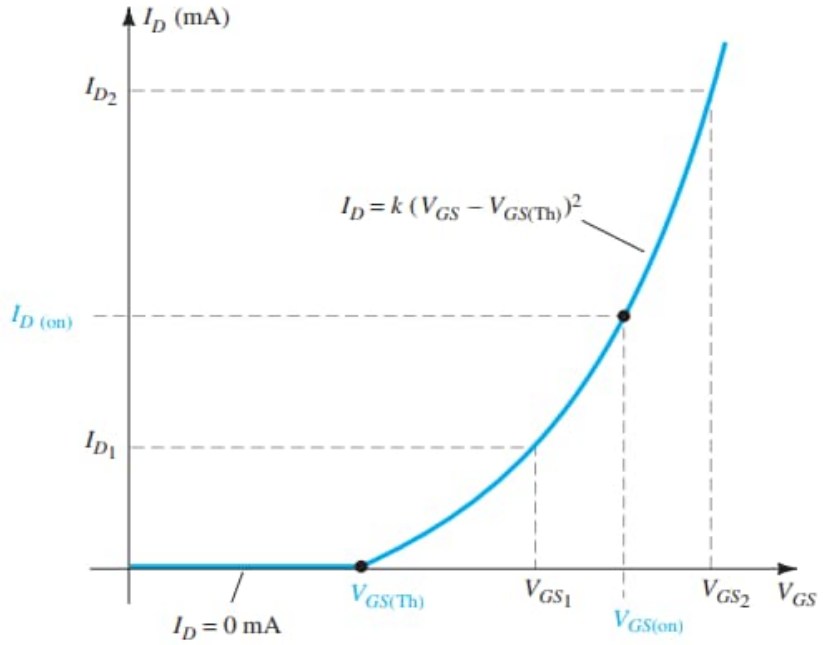


Fig.3

This curve is obtained after knowing the parameter k, which can be found using the equation:

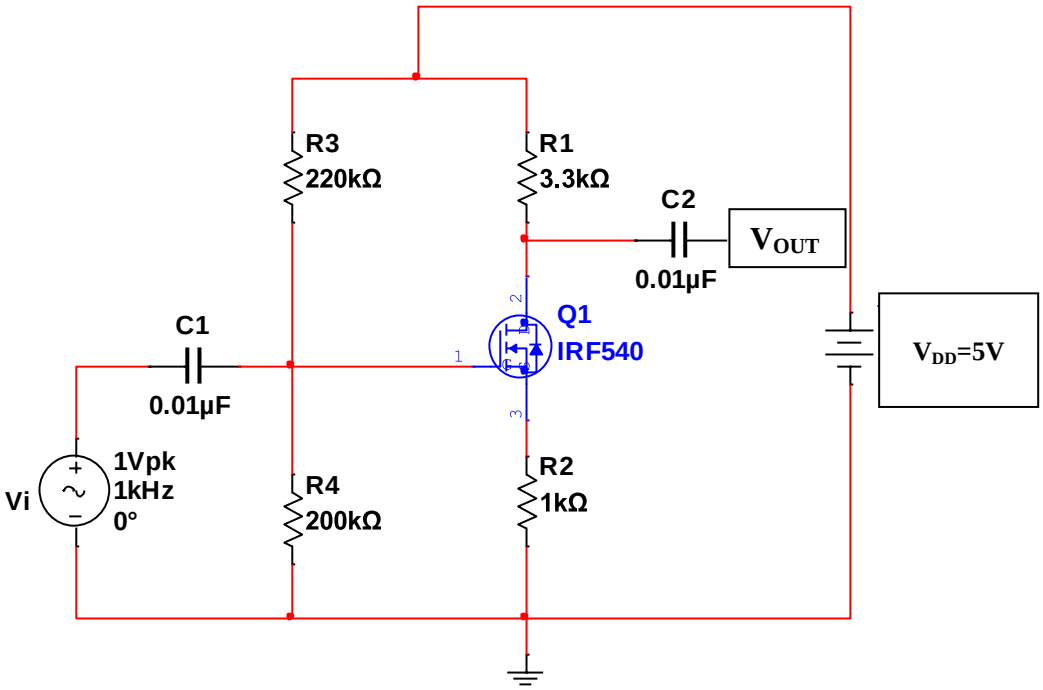
$$k = \frac{I_{D(on)}}{\left(V_{GS(on)} - V_{th}\right)^2}$$

The drain current and gate voltages needed to calculate k, are taken from the data sheet of the MOSFET being used.

Equipment and Component:

Serial no.	Component Details	Specifications	Quantity
1	MOSFET	IRF540n	1 piece
2	Resistor	1kΩ, 3.3kΩ, 200kΩ, 220kΩ	1 piece each
3	Capacitors	0.01 uF	2 pieces
4	Function Generator Oscilloscope		1 unit 1unit
5	DC Power Supply		1 unit
6	Digital Multimeter		1 unit
7	Chords and Wires		1 unit

Experimental Circuit Diagram:



Procedure:

1. Implement the circuit on the bread board
2. Connect the channels(CH-1 and CH-2) of the oscilloscope with input and output terminals
3. Set the input voltage Vi=1Vpk and frequency = 1KHz by using Function generator.
4. Observe the input and output signals on the oscilloscope screen. If faithful output is found, measure V_i , Vo and phase difference.
5. Calculate the voltage gain.

Simulation:

By simulating the experimental circuit collect the value of Vi and Vo. Calculate voltage gain and copare the result with practical data.